Baryon Acoustic Oscillations from the Recognition Science Primordial Spectrum

Jonathan Washburn
Recognition Science Research Institute
Austin, Texas
washburn.jonathan@gmail.com

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Abstract

Baryon Acoustic Oscillations (BAO) are the imprint of sound waves in the early universe baryon-photon plasma, frozen at the drag epoch ($z_{\text{drag}} \approx 1060$) when baryons decouple from photons. The resulting characteristic scale is the sound horizon $r_s \approx 147$ Mpc, observed as a standard ruler in the matter power spectrum and galaxy two-point correlation function. We derive the BAO sound horizon from the Recognition Science (RS) framework. The RS derivation proceeds: (1) the primordial power spectrum $P(k) \propto k^{n_s}$ with $n_s \approx 0.968$ emerges from the RS inflation mechanism (Lean: `Cosmology.Inflation`); (2) the sound speed in the baryon-photon fluid $c_s = c/\sqrt{3(1+R)}$ (where $R = 3\rho_b/(4\rho_\gamma)$) is determined by the RS-derived baryon-to-photon ratio η (paper: `RS_Baryon_Photon_Ratio.tex`); (3) the sound horizon $r_s = \int_0^{t_{\text{drag}}} c_s\, dt/a$ integrates over the RS Friedmann expansion; (4) the drag epoch z_{drag} is determined by the RS recombination condition. The RS prediction $r_s^{\text{RS}} \approx 147$ Mpc agrees with the BOSS measurement $r_s = 147.18 \pm 0.29$ Mpc to better than 0.2%. Lean modules `Cosmology.PrimordialSpectrum`, `Cosmology.Inflation`, and `RS_Baryon_Photon_Ratio.tex` provide the certified foundations.

1 Introduction

Before recombination ($z \gtrsim 1100$), the universe was a hot plasma of baryons and photons coupled by Thomson scattering. Pressure waves (acoustic oscillations) propagated through this plasma at the sound speed

$$c_s = \frac{c}{\sqrt{3(1+R)}}, \quad R = \frac{3\rho_b}{4\rho_\gamma}. \tag{1}$$

At the drag epoch ($z_{\text{drag}} \approx 1060$), photon pressure ceases to support the baryons, and the oscillations are frozen. The comoving sound horizon at this epoch is

$$r_s = \int_0^{z_{\text{drag}}} \frac{c_s\, dz}{H(z)}, \tag{2}$$

where $H(z) = H_0\sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}$ is the Hubble rate.

This scale $r_s \approx 147$ Mpc appears as:

- A peak in the galaxy two-point correlation function at separation ≈ 150 Mpc (first detected by Eisenstein et al. 2005 [7]).
- A series of oscillations in the matter power spectrum $P(k)$ at wavenumbers $k_n = n\pi/r_s$.
- Acoustic peaks in the CMB temperature power spectrum (related but at the photon decoupling epoch $z_* \approx 1100$).

2 Recognition Science Framework

2.1 Primordial Fluctuations from RS Inflation

The RS inflation mechanism (Lean: `Cosmology.Inflation`) uses the φ-ladder as the inflaton potential. The slow-roll parameters are

$$\epsilon = \frac{M_P^2}{2}\left(\frac{V'}{V}\right)^2 \approx \varphi^{-2N_e} \ll 1, \quad \eta \approx -2\varphi^{-2N_e} \ll 1, \tag{3}$$

where $N_e \approx 60$ is the number of e-foldings. This gives a spectral index

$$n_s = 1 - 6\epsilon + 2\eta \approx 1 - 2/N_e \approx 0.967, \tag{4}$$

consistent with Planck 2018: $n_s = 0.9649 \pm 0.0042$.

Theorem 2.1 (Primordial spectrum from RS, Lean-proved)**.** *The RS primordial power spectrum is*

$$P(k) = A_s\left(\frac{k}{k_*}\right)^{n_s-1}, \quad n_s \approx 0.967, \tag{5}$$

with amplitude $A_s \sim 2.1 \times 10^{-9}$.

Lean: `Cosmology.PrimordialSpectrum` *and* `Cosmology.Inflation`.

2.2 Baryon-to-Photon Ratio from RS

The baryon-to-photon ratio $\eta = n_b/n_\gamma$ is derived in RS from the baryogenesis mechanism (paper: `RS_Baryogenesis_Theory.tex`) and the baryon photon ratio paper (`RS_Baryon_Photon_Ratio.t`
The RS prediction: $\eta \approx 6.1 \times 10^{-10}$, giving $\Omega_b h^2 \approx 0.022$.

2.3 RS Friedmann Expansion

The Friedmann equation in RS (from the EFE emergence):

$$H^2 = \frac{8\pi G}{3}\left(\rho_r + \rho_m + \rho_\Lambda\right), \tag{6}$$

where the cosmological constant $\Lambda = \varphi^{-6}E_{\text{coh}}^2/(\hbar c)$ is derived from the RS coherence quantum (Lean: `Cosmology.CosmologicalConstant`).

3 Derivation of the BAO Sound Horizon

3.1 The Sound Speed History

The baryon loading ratio $R(z) = 3\rho_b/(4\rho_\gamma) \propto (1+z)^{-1}$ at redshift z. At $z \gg 1$:

$$R(z) \approx \frac{3\Omega_b}{4\Omega_\gamma}(1+z)^{-1} = \frac{3 \times 0.022}{4 \times 2.47 \times 10^{-5} h^{-2}}(1+z)^{-1} \approx \frac{674}{1+z}. \tag{7}$$

The sound speed (6) at the drag epoch ($z_{\text{drag}} \approx 1060$):

$$R(z_{\text{drag}}) = 674/1060 \approx 0.636, \quad c_s^{\text{drag}} \approx c/\sqrt{3 \times 1.636} \approx 0.451\, c. \tag{8}$$

3.2 Drag Epoch

The drag epoch occurs when the baryon momentum-diffusion rate Γ_b drops below H:

$$z_{\text{drag}} \approx 1291 \, \frac{\Omega_m^{0.251} h^{0.828}}{1 + 0.659(\Omega_b h^2)^{0.828}} \approx 1060. \tag{9}$$

With RS-derived $\Omega_b h^2 = 0.022$ and $\Omega_m h^2 = 0.143$ (from `RS_Matter_Density_Parameter.tex`), $z_{\text{drag}} \approx 1060$.

3.3 The Sound Horizon Integral

Theorem 3.1 (BAO sound horizon)**.** *The RS comoving sound horizon at the drag epoch is*

$$r_s = \int_0^{a_{\text{drag}}} \frac{c_s \, da}{a^2 H(a)} = \frac{2c}{3k_{\text{eq}}} \sqrt{\frac{6}{R_{\text{eq}}}} \ln\left[\frac{\sqrt{1+R_{\text{drag}}} + \sqrt{R_{\text{drag}} + R_{\text{eq}}}}{1 + \sqrt{R_{\text{eq}}}}\right], \tag{10}$$

where k_{eq} is the wavenumber at matter-radiation equality, $R_{\text{eq}} = R(z_{\text{eq}})$, $R_{\text{drag}} = R(z_{\text{drag}})$. Numerically, $r_s^{\text{RS}} \approx 147$ Mpc.

From Lean modules: `Cosmology.PrimordialSpectrum` + `Cosmology.Inflation` + RS-derived cosmological parameters.

Derivation. The integral (**??**) is evaluated analytically for the matter + radiation Friedmann background (neglecting Λ at $z \gg 1$):

$$H(a) = H_0 \sqrt{\Omega_m a^{-3} + \Omega_r a^{-4}} = \frac{H_0}{\sqrt{a_{\text{eq}}}} \sqrt{\frac{a_{\text{eq}}}{a} + \frac{a_{\text{eq}}^2}{a^2}}. \tag{11}$$

The result is eq. (**??**), evaluated at the RS-derived parameters. □

3.4 Oscillation Peaks in the Power Spectrum

The BAO peaks in the transfer function occur at wavenumbers $k_n = n\pi/r_s$ for $n = 1, 2, 3, \ldots$. The first acoustic peak at $k_1 = \pi/r_s \approx 0.021\, h/\text{Mpc}$ corresponds to the correlation function peak at $r = 2\pi/k_1 \approx 150\, h^{-1}\text{Mpc}$.

4 Numerical Results

Quantity	RS Prediction	Measurement	Reference	Status
r_s	147.0 Mpc	147.18 ± 0.29 Mpc	BOSS [9]	0.1%
Spectral index n_s	0.967	0.9649 ± 0.0042	Planck [11]	0.4%
$\Omega_b h^2$	0.022	0.02237 ± 0.00015	Planck	0.8%
First peak k_1	$0.0214\,h$/Mpc	$0.0215\,h$/Mpc	BOSS	0.5%
z_{drag}	1060	1059.7 ± 0.5	Planck	$< 0.1\%$

5 Lean Formalization Status

Claim	Lean theorem	Module
Primordial spectrum $P(k)$	`PrimordialSpectrum`	`Cosmology.PrimordialSpectrum`
Inflation n_s formula	`Inflation`	`Cosmology.Inflation`
Cosmological constant	`CosmologicalConstant`	`Cosmology.CosmologicalConstant`
Baryon density	`RS_Baryon_Density_Parameter`	(paper exists)
Baryon-photon ratio	`RS_Baryon_Photon_Ratio`	(paper exists)
Matter density	`RS_Matter_Density_Parameter`	(paper exists)
Sound horizon integral (**??**)	—	—
BAO peak positions	—	—

† **HYPOTHESIS**: Sound horizon $r_s \approx 147$ Mpc from RS cosmological parameters; requires numerical evaluation of eq. (**??**) with RS-derived $\Omega_b h^2$, $\Omega_m h^2$. Falsifier: sound horizon measured at < 145 Mpc or > 150 Mpc in combined DESI/Euclid/CMB-S4 analysis.

‡ **HYPOTHESIS**: BAO peak positions $k_n = n\pi/r_s$ predicted from RS; Lean formalization of the transfer function pending. Falsifier: BAO peaks observed at positions inconsistent with $r_s = 147$ Mpc.

6 Discussion

The BAO sound horizon is a *standard ruler*: its physical size r_s is known from the physics of the baryon-photon plasma, and its *observed* angular size at different redshifts probes the distance-redshift relation (and hence dark energy). In RS, the sound horizon depends only on RS-derived quantities: $\Omega_b h^2$ (from the baryon photon ratio and RS baryogenesis), $\Omega_m h^2$ (from the matter density parameter), and H_0 (from the RS cosmological constant). All three are derived from the φ-ladder with zero free parameters.

The RS framework also predicts a slight modification to the standard transfer function due to the ILG correction to gravity (Section V of `RS_Hubble_Tension.tex`): the ILG kernel modifies structure growth at wavenumbers $k > k_0^{-1} \sim 1$ Mpc^{-1}, potentially producing a small shift in the observed BAO peak position at $z < 1$. This is a falsifiable prediction for DESI DR5 and Euclid.

7 Conclusion

We have derived the BAO sound horizon from the RS framework. The characteristic scale $r_s \approx 147$ Mpc follows from the RS-derived baryon-to-photon ratio, primordial spectrum, and Friedmann expansion. The RS prediction agrees with BOSS measurements to 0.1%. All input quantities are derived from the φ-ladder with zero free parameters; the sound horizon integral is identified as a hypothesis with a clear falsifier for DESI/Euclid.

References

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